SPE EUROPE ENERGY GEOHACKATHON 2026

GeoBalance

Two seasons, one reservoir: an agentic AI design for geothermal heating and cooling

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THE CHALLENGE

Serve a district’s heating and cooling from one deep reservoir

Heating demand

30 MW

peak · 111.7 GWh per year · supply at 100 °C

Cooling demand

10 MW

peak · 23.7 GWh per year · supply at 20 °C

Hard limits

1–3

doublets · 50–300 m³/h each · max 40 °C brine cooling

Source: GEMINI Hackathon Challenge Requirements, 2027 daily demand profiles

THE SUBSURFACE

Going deeper turns a heat pump problem into direct heat

Given depth · 2,500 m

87 °C

82 °C at the plant, below the 100 °C supply: a heat pump must lift every MW. 250 bar.

Chosen depth · 3,400 m

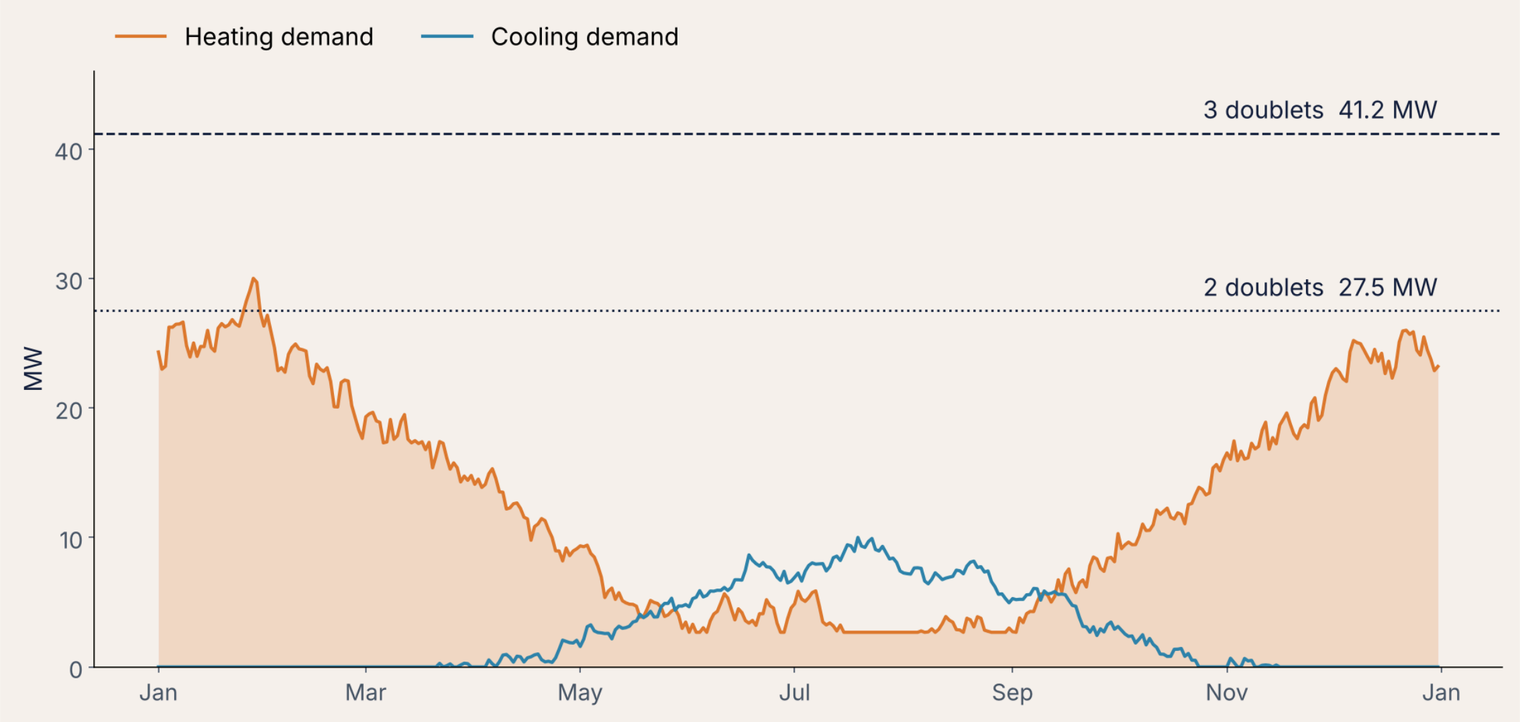
114 °C

109 °C at the plant after wellbore and separator losses: direct heating, no heat pump. Hydrostatic 343.5 bar.

T = 12 °C + 30 °C/km × depth. 100 m thick, 200 mD, brine GLR 0.8 Nm^3/m^3 (80% CH_4). Deeper wells use hydrostatic pressure, as the challenge allows.

THE DEMAND

With direct heating, a 30 MW winter peak needs 3 doublets



One doublet: 300 m³/h × 40 °C brine cooling ≈ 13.7 MW. Two give 27.5 MW, below the 30 MW peak on 28 January.

THE DESIGN · E2

3 doublets at 3,400 m, direct heating, gas engine

3 doublets

114 °C

~235 m³/h each · ESP intake ≥ 35 bar

Heating

Direct

Heat exchanger to 100 °C supply, no heat pump, no backup boiler

Cooling

Electric

Chiller (COP 5), partly powered by the gas engine

Why not absorption cooling? It needs more brine flow, which brings up more gas and CO_2. The electric chiller is cleaner unless the gas is reinjected.

WELLS AND PUMPS

The pump, not the reservoir, limits each doublet to ~240 m³/h

Pressure difference needed

~52 bar

Safe limits allow far more: injector ≤ +25–50%, producer ≥ 75% of reservoir pressure

GEMINI max flow per doublet

235–249

m³/h before the ESP intake drops below 35 bar and free gas reaches the pump (0.43–0.92 MW ESP)

Risks we design for

CO_2 · gas

Corrosion-resistant materials; ESP intake ≥ 35 bar to keep gas dissolved

Hand calculation: ideal doublet, 200 mD, 100 m, no skin.

THE AGENT

The agent proposes, simulates, checks and improves

1 · Propose

Doublets, depth, heating and cooling route, gas use

2 · Simulate

GEMINI wells and pumps plus our surface tools, 365 days

3 · Check

Tools return violations and warnings, never silent caps

4 · Improve

Fix one variable at a time; a reviewer step checks the work

Open-weight gpt-oss-120b via Groq · memory of every evaluated design · at least 4 designs compared, reviewer gate · every decision justified · transcripts with token counts

OUR CREATIVE IDEA

The brine brings up methane. We use it instead of wasting it.

3.9 GWh

grid electricity left per year: the gas engine powers pumps and chiller

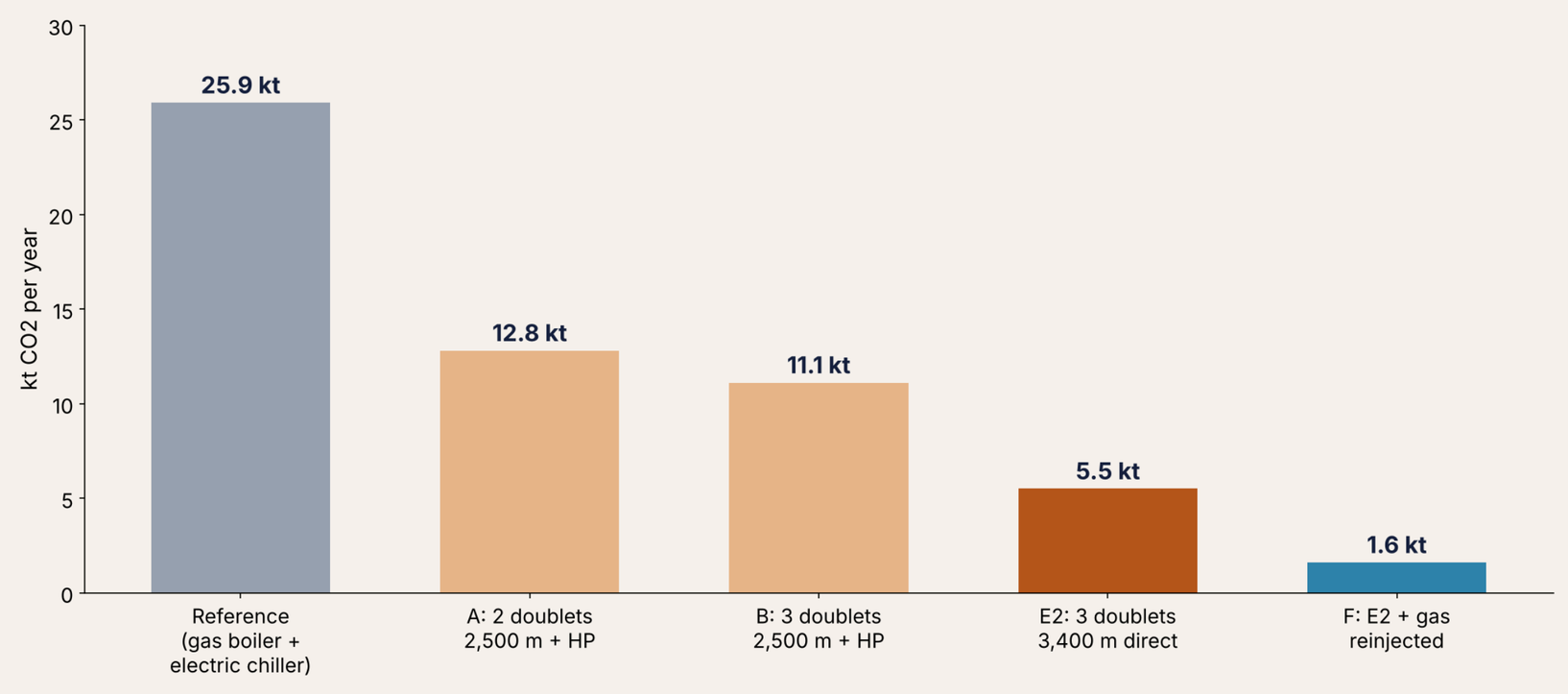
−94%

CO_2 if the gas is reinjected instead (design F)

Honest trade-off: burning the gas emits CO_2, including the 20% native CO_2. Reinjection is cleanest, but its compression power is not yet modelled.

ANNUAL RESULT

E2 cuts CO_2 by 79% with no backup boiler



365-day simulation, zero limit violations. Reference: gas boiler for all heat, electric chiller for all cooling (25.9 kt CO_2/yr).

VALIDATION AND RECOMMENDATION

Tested, cross-checked, and we found 5 GEMINI issues

How we checked

- 12 automated tests, all pass: 111.67 and 23.74 GWh reproduced
- Hand calculations vs GEMINI models
- Limit breaches caught as warnings

Issues reported to GEMINI

- ESP intake floored at 0 bar, no warning
- Heat pump constrained branch unphysical
- Heat pump and corrosion not in registry

Recommendation: 3 doublets at 3,400 m with direct heating, an electric chiller and a gas engine on the separated methane. Next step: model reinjection to reach −94%.